SES 494 Honors Project (SPR ‘26)

F10.7 Index and Solar Flux

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The F10.7 Index and Solar Flux

Abstract

The F10.7 Index is the index of the radio flux emitted from the sun at the 10.7 cm wavelength, or the 2800 MHz frequency. This critical measurement is necessary for calculations modeling and analyzing orbital lifetime for satellites and spacecraft (2026). F10.7 is also used in predicting solar activity to help prevent ion interference in ground-based technology. In this paper, we will begin by discussing the history of the F10.7 index, the telescopes that are used to collect data, how and where to find the data. Then we will move onto why the F10.7 index is important, and finally, look at plotted sample data and discuss what it represents.

1. Introduction: Brief History of the F10.7 Index

In the long history of astronomy and astrophysics, the F10.7 index is fairly recent, having only been discovered during WWII by accident. Increased receiver noise, called sun strobes, were seen on radar displays when dishes were pointed towards the sun, a pattern noticed then reported by George Clark Southworth, a prominent radio engineer. At first, it was thought to be, "random noise in the output of a conventional double-detection radio receiver", but Southworth later found it to be in line with results that would come from applying the black body radiation theory. This led Southworth and his team to determine it was radio signals from the sun (1945).

Unfortunately, because this was taking place in the middle of a world war, and was related to military radar development, this information was kept restricted. Afterwards, when these studies were allowed to be publicly published, and the scientific community was swimming in new, cutting-edge technology, a new influx of interest in radio astronomy brought in fresh minds and federal funding. The Canadian National Research Council used old parts from the war to build out a radio telescope just south of Ottawa that would scan at the reported 10.7 cm wavelength. In 1947, and with physicist Arthur Covington at the head of the research team, the National Research Council quickly realized that the solar flux measured changed daily, and corresponded with recorded disk brightness temperatures. Interestingly enough, this Lehaney and Yabsley in 1949. Spurred by this discovery, the National Research Council began recording daily. Due to urban expansion from the nearby capital, the telescope began experiencing increased and debilitating signal interference. Unable to continue gathering usable data, two more telescopes were built— the first at the Algonquin Radio Observatory (ARO), and the second at the Dominion Radio Astrophysical Observatory (DRAO). Both telescopes were used to continue recording solar measurements for next seventeen years, until Covington's retirement in 1979, when the ARO 10.7 cm telescope was closed. Observations were moved to DRAO, where they still remain open and gathering data today (2013).

Today, the Solar Radio Monitoring Program is supported not only by the National Research Council, but also by Natural Resources Canada and the Canadian Space Agency. There are two current instruments that record, and are open-source data (2015). All data collected by the National Research Council can be found on the Canadian Space Weather website, maintained by the Government of Canada (2015). Immediate delivery of data once recorded can be faxed or sent directly to email upon request.

2. F10.7 Index Definition

As mentioned in the abstract, the F10.7 index is an integral part of space weather science. It is the longest running record of solar activity, and is one of the few solar indexes that can be reliably measured from the Earth's surface, as well as recorded in any type of weather (2018). This index is recorded in solar flux units (s.f.u.). 1 s.f.u. is defined as $10^{-22}\ W\ m^{-2}\ Hz^{-1}$.

3. Data and Data Breakdown

Data for F10.7 is available from 1949, however, data from before October 28, 2004 is no longer available directly from the Canadian space weather website. Instead, direct download grants all data from October 28, 2004 to present, in a .txt format (2025). Previous data can be requested via email. All data is recorded daily between 2-3 times a day, and recorded in a format of date, time, Julian Calendar date, the Carrington Number, the observed flux, the adjusted flux, and the USRI Series D (2025).

The Carrington number is the number of rotations of the sun since 9th November, 1853. The observed flux is the number measured directly from the ground-based radio telescope, modulated by the level of solar activity, and the seasonal distance between the Earth and the Sun. This flux number is used when studying terrestrial data. The next number recorded is the adjusted flux, which gives the average distance between the Earth and Sun. Because this data needs to cooperate with other flux data at different frequencies for research, it needs to be modified by a scaling factor. The USRI Series D is the adjusted value, multiplied by 0.9.

4. Plotted Data and Explanation

For this paper, we will be looking at flux data between 2004 and present, taking every 90 samples recorded, then plotting them. This number was chosen to take a data point from each month, with some overlap, mostly for readability. Then, we will look at data averaged from every month, to compare. Then, we have plotted all data from 2004 to present. Data was pulled from the Canadian Space Weather website on 24 February, 2026. The full data contains 23,333 samples. First, we'll look at the sampled data from every 90 days.

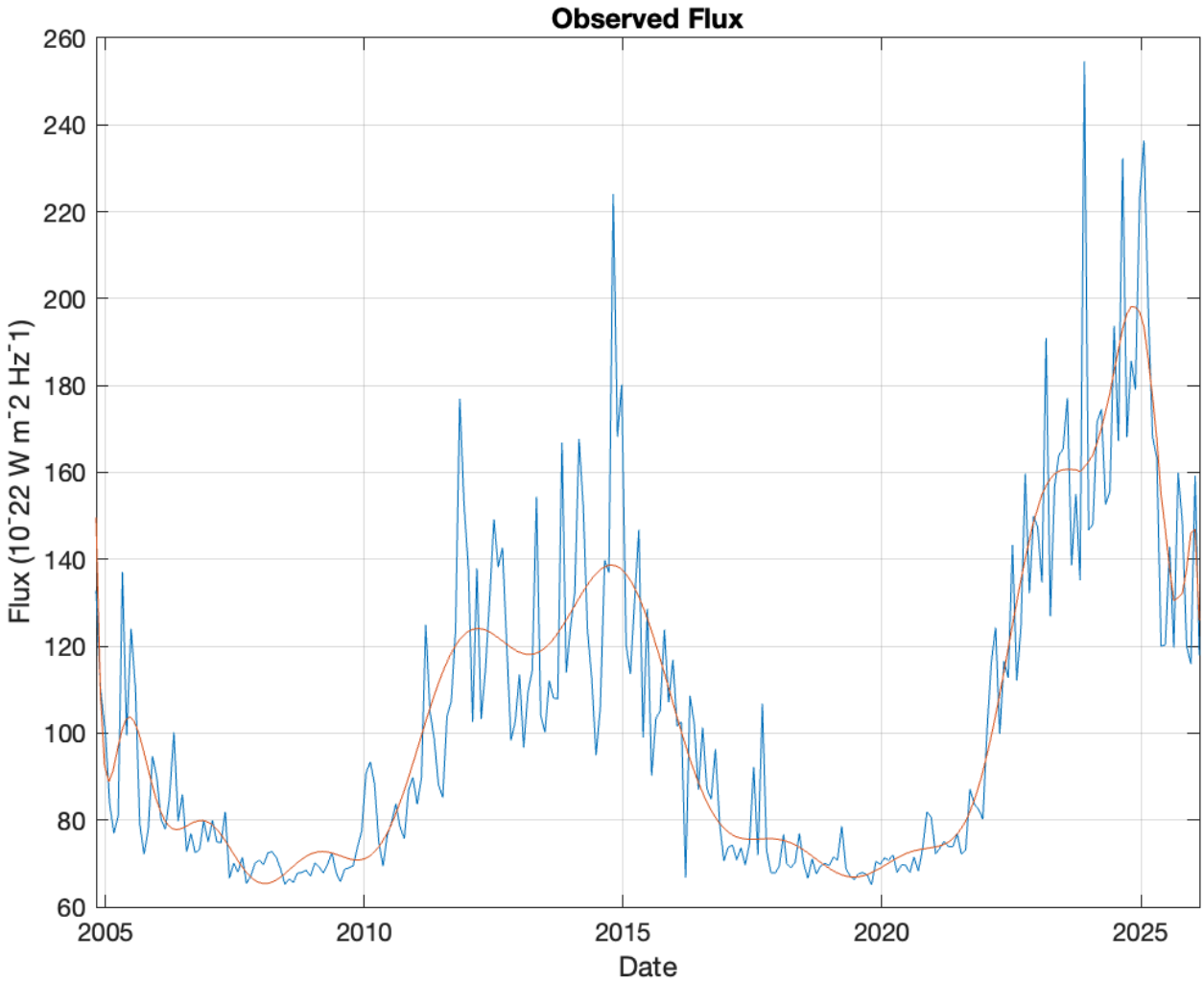

Figure 1. Plotted observed solar flux from October 28, 2004 to February 16, 2026. An orange line of best fit is overlaid on the data.

In this figure, you can clearly see the solar cycle. Starting from the left, we can see the end of Solar Cycle 23, as it goes into Solar Cycle 24. Beginning in December 2019, Solar Cycle 25 begins, and is expected to continue until around 2030.

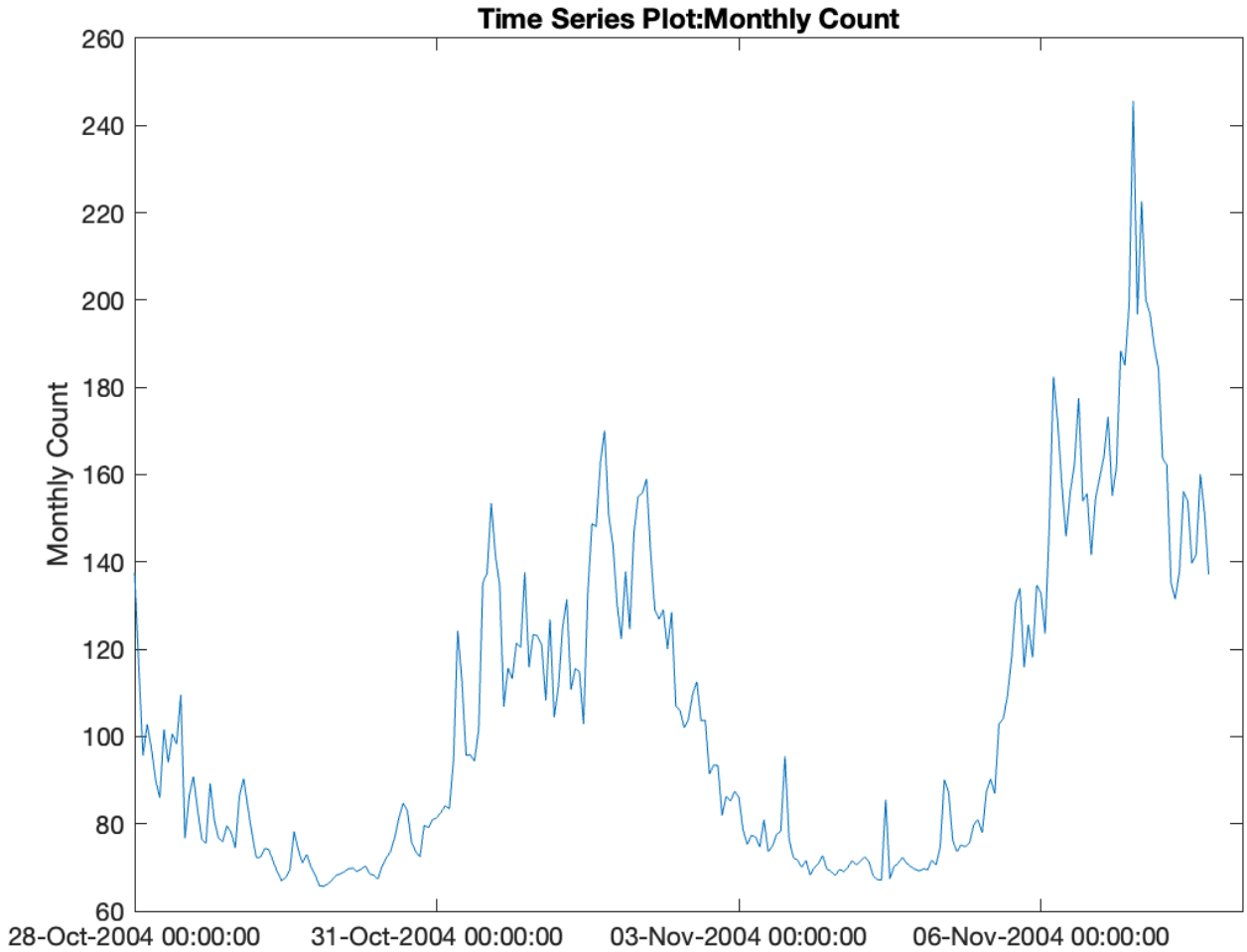

Figure 2. Plotted observed solar flux from October 28, 2004 to February 16, 2026, averaging all data from every month.

In this figure, all monthly data has been added and averaged. Because all data is averaged, the graph is more readable, but solar flares are not shown as sharply as they will be in Figure 3.

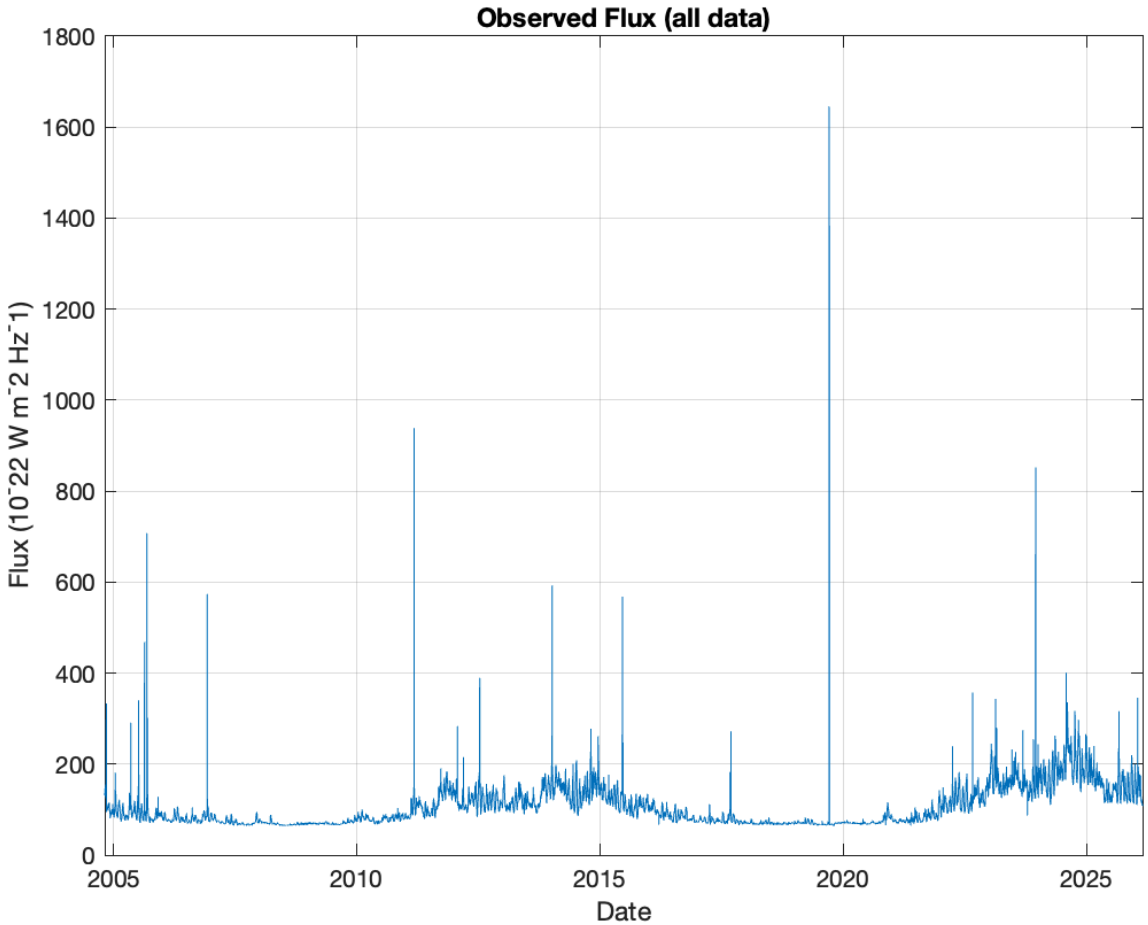

Figure 3. Plotted observed solar flux from October 28, 2004 to February 23, 2026. 23,333 points of data are present.

In Figure 3, we can see all data from 2024 to present plotted. This data is much less readable, because we can see the spikes, such as the spike on 17th September, 2019, or on 7th March, 2011. The data point from 7th March matches up with a Coronal Mass Ejection (CME) near sunspot 1164 (2011). The data point on 17 September matches with a solar wind event originating in a southern hole from the sun's atmosphere (2019). This had the effect of adding to arctic auroras, meaning the solar wind could be visibly seen from Earth.

One of the figures the F10.7 index helps measure is sunspots during the solar cycle. A solar cycle is about 11 years long, and since the beginning of the F10.7 indexing, there have been over seven solar cycles worth of sunspots and solar activity calculated and recorded (2026). These numbers can help predict future solar flares, an important factor in maintaining ground-based technology. A solar flare can send bursts of ions through the atmosphere, and in a world fatally reliant on technology, a single bit flip caused by solar flares could be as bad as a computer suddenly telling a commercial aircraft the data its reading is now different kind of data (2008).

Since the F10.7 index measures solar flux, it is also used in calculating orbital lifetime for orbiting craft around Earth. This is because solar flux, along with geomagnetics, are the prime controlling factors in atmospheric behavior and activity. Using both of these quantities, we can

use them to calculate the density of the atmosphere, at the location and altitude of a satellite. The history of this data is important, because past data can be used to predict behaviors from our Sun, and so an updated version of what behaviors the star will probably exhibit, the mean behavior, are calculated and published by NASA and NOAA. Since the lifetime of a satellite in orbit depends on the drag of the atmosphere, especially in low-Earth orbit (LEO), this calculation is vital to maintain satellite up-time and physical life.

5. Conclusion

Radio astronomy is incredibly important for our everyday life, and not just for space weather science. At the core of a lot of our daily lives is the F10.7 index, and the predictions it helps calculate. This paper explained its history, how the data is collected, and what the data is used for.

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